

Lucas Ferreira

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Education

Georgia Institute of Technology - B.S. in Civil Engineering

August 2018-May 2021

Professional Experience

NoTraffic, Atlanta, Georgia

Signal Optimization Engineer

November 2024-Present

- Lead end-to-end deployment of NoTraffic's AI-driven adaptive signal control algorithms across major municipalities, including Phoenix, Topeka, Missouri City, Naples, Edmonton, and Montreal.
- Configured and tuned AI traffic signal control models across national corridors, using live performance data to iteratively adjust algorithm parameters, reducing intersection delay by up to 30% and travel time by 10%.
- Primary technical liaison between overseas R&D and U.S. field operations - translating ambiguous, real-world traffic signal anomalies into concrete AI model tuning specs and product improvements.
- Architected a multi-agent AI system (Claude Code and custom Metabase MCP server) to automate algorithm configuration workflows and client report generation, cutting a 12-hour manual process to under 1 hour.

Q-Free ASA, Atlanta, Georgia

Support Engineer

November 2021-November 2024

- Owned product support and solution design for the Q-Free Urban platform, helping secure statewide enterprise deployments for GDOT and NCDOT.
- Directed technical UAT and edge-case controller hardware validation for major municipal rollouts (City of Laval, City of Dallas, NCDOT), standardizing testing scripts to slash product implementation timelines.
- Served as the technical lead on-site for advanced adaptive control system deployments in Conway, AR and Boulder, CO, managing edge-case networking environments and hardware implementation.

Kimley-Horn, Atlanta, Georgia

Civil Engineering Analyst

June 2021-November 2021

- Analyzed traffic flow and congestion patterns using Excel, Python, and MySQL, leading to increased traffic flow, and reduced greenhouse gas emissions for municipalities in the state of Georgia,

Leadership & Community Involvement

Beta Iota Alumni Board, Atlanta, Georgia

President

May 2026-Present

- Promoted to president of the non-profit organization. I oversee endowment asset allocations, and provide strategic advisory to active undergraduate organizational leaders.

Treasurer

January 2023-May 2026

- Manage chapter alumni finances: budgeting, tax filing, and financial reporting.

Society of St. Vince De Paul, Atlanta, Georgia

Caseworker

January 2026-Present

- Evaluate and assist families and individuals in need through St. Jude's Parish. Conduct case assessments, connect clients with resources, and provide direct support to community members facing hardship.

Projects

ATMS-Lite: Real-Time Traffic Signal Management System | Python, FastAPI, React, Docker, SNMP

- Built a locally-hosted traffic management system modeling the intersections in Atlanta polling live telemetry at ~5 Hz over SNMP from a physical hardware controller; containerizing a multi-intersection test environment via Docker Compose.

Lulu: Local AI Desk Companion & Voice Pipeline | ESP32-S3, Python, WebSockets, Ollama, PlatformIO

- Designed and built an embedded hardware AI companion with a fully local, cloud-free voice pipeline (Whisper STT, Ollama LLM, Piper TTS) over a custom low-latency WebSocket backend.

Skills

- **Programming & Scripting:** Python (Advanced), Bash, JavaScript, React 19, FastAPI, Tailwind v4, REST APIs

- **Cloud & Infrastructure:** Docker, PostgreSQL, MySQL, SQLite, Linux (Ubuntu), Git, Microsoft Azure, CI/CD (Jenkins)
- **Languages:** Portuguese (Native), Spanish (Fluent)